15 (a) Factorise $a^2 - b^2$

×	a	$-b$
a	a^2	$-ab$
b	ab	$-b^2$

The +ab and −ab cancel.

1 mark
Difference of two squares

$(a - b)(a + b)$

(1)

(b) Hence, or otherwise, simplify fully $(x^2 + 4)^2 - (x^2 - 2)^2$

Use $a = x^2 + 4$ and $b = x^2 - 2$.

1 mark
Identify the two squares

$a - b = (x^2 + 4) - (x^2 - 2) = 6$

$a + b = (x^2 + 4) + (x^2 - 2) = 2x^2 + 2$

1 mark
Multiply their difference by their sum

$6(2x^2 + 2) = 12x^2 + 12$

1 mark
Correct simplified expression

$12x^2 + 12$

(3)

(Total for Question 15 is 4 marks)

16 There are only red counters, blue counters and purple counters in a bag.
The ratio of the number of red counters to the number of blue counters is 3 : 17

Sam takes at random a counter from the bag.
The probability that the counter is purple is 0.2

Work out the probability that Sam takes a red counter.

(Total for Question 16 is 3 marks)

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